(4) prove the following

(1) $L[e^{at}] = \frac{1}{s-a}$

(2) $L[\cos at] = \frac{s}{s^2+a^2}$

(3) $L[\sin at] = \frac{a}{s^2+a^2}$

(4) $L[\cosh at] = \frac{s}{s^2-a^2}$

(5) $L[\sinh at] = \frac{a}{s^2-a^2}$

(6) $L(t^n) = \frac{n!}{s^{n+1}}$

~~(7) $L(\sin 2t)$~~

II Find the laplace Transform of the following functions

(11) $L[\sin^2 t]$

(12) $L[\sin 3t \cos t]$

(13) $L[\sin t \sin 2t \sin 3t]$

(14) $L[1+e^{-3t}-5e^{4t}]$

(15) $L[(1+t)^2]$

(16) find the laplace transform of $f(t) = \begin{cases} e^t & 0<t<1 \\ 0 & t>1 \end{cases}$

(17) $L\left[\frac{1}{\sqrt{t}} + t^{3/2}\right]$

(18) $L[e^{3t} \sin^2 4t]$

(19) $L\left[\frac{1-e^{2t}}{t}\right]$

(20) $L\left[\frac{e^{-at}-e^{-bt}}{t}\right]$

(21) $L\left[\frac{\sin^2 t}{t}\right]$

(22) find the Laplace transform of $G(T)$, where

$$G(T) = \begin{cases} \cos\left(t - \frac{2\pi}{3}\right) & \text{if } t > \frac{2\pi}{3} \\ 0 & \text{if } t < \frac{2\pi}{3} \end{cases}$$

(23) Assuming $L(\sin t)$. Find $L(\sin 2t)$ & $L(\sin t/2)$

(24) find $L\left(e^{-t}\int_0^t t\cos t\,dt\right)$

(25) find $L\left(\int_0^t e^{-t}\sinh 2t\,dt\right)$

(26) Find the Laplace Transform of the square wave given by

$$f(t) = \begin{cases} E & \text{for } 0 < t < a/2 \\ -E & \text{for } a/2 < t < a \end{cases}$$

(27) find the laplace transform of $f(t) = \begin{cases} \sin t & \text{in } 0 < t < \pi \\ 0 & \text{in } \pi < t < 2\pi \end{cases}$

& $f(t+2\pi) = f(t)$

(28) if $L(f(t)) = \frac{1}{s(s+a)}$ find $\lim_{t\to 0} f(t)$ & $\lim_{t\to\infty} f(t)$

(29) verify initial & final value theorem for the function $f(t) = e^{-2t}\cos 3t$

Inverse Laplace transform

Find the inverse Laplace transform of the following functions

(1) $L^{-1}\left[\frac{1}{s^2}+\frac{1}{s+4}+\frac{1}{s^2+4}+\frac{s}{s^2-9}\right]$

(2) $L^{-1}\left[\frac{s-3}{(s-3)^2+4}\right]$

(3) $L^{-1}\left[\frac{3s-4}{s^2-8s+65}\right]$

(4) if $L^{-1}\left[\frac{s^2-1}{(s^2+1)^2}\right] = t\cos t$, then find $L^{-1}\left[\frac{9s^2-1}{(9s^2+1)^2}\right]$

(5) $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$

(6) $L^{-1}\left[\tan^{-1}(a/s)+\cot^{-1}(s/b)\right]$

(7) $L^{-1}\left[\log\frac{s(s^2+1)(s-4)^2}{(s^2-9)(s^2+4)}\right]$

(8) $L^{-1}\left(\frac{54}{s^3(s-3)}\right)$

(9) $L^{-1}\left(\frac{1}{s(s^2+6s+13)}\right)$

(10) $L^{-1}\left(\frac{e^{-s}}{(s+1)^{5/2}}\right)$

(11) $L^{-1}\left[\frac{e^{-as}}{s^2}\right]$, $a>0$

(12) $L^{-1}\left[\log\frac{s+3}{s(s-2)}\right]$

(13) $L^{-1}\left[\frac{2s^2+5s+2}{(s-3)^4}\right]$

(14) use convolution theorem to find $L^{-1}\left(\frac{1}{(s+a)(s+b)}\right)$

Find inverse Laplace Transform using partial fractions

(15) $L^{-1}\left(\frac{1}{(s+1)(s+3)}\right)$

(16) $L^{-1}\left(\frac{5s^2-15s-11}{(s+1)(s-2)^3}\right)$

(17) $L^{-1}\left(\frac{2s^2+5s+2}{(s-3)^4}\right)$

(18) $L^{-1}\left(\frac{1-s}{(s+1)^2(s^2+4s+13)}\right)$

(19) $L^{-1}\left(\frac{2s^2-6s+5}{s^3-6s^2+11s-6}\right)$

(20) $L^{-1}\left[\frac{1}{s(s+1)^3}\right]$

(21) $L^{-1}\left[\frac{1}{(s+1)^2(s^2+4)}\right]$

(22) $L^{-1}\left[\frac{1}{(p+2)^2(p-2)}\right]$

(23) $L^{-1}\left[\frac{1}{(s+1)(s^2+2s+2)}\right]$

(24) $L^{-1}\left[\frac{86s-78}{(s+3)(s-4)(5s-1)}\right]$

(25) $L^{-1}\left[\frac{1}{(s+1)(s^2+1)}\right]$

Applications of Laplace transform

III using Laplace transforms, solve the following

(1) $y'' - 4y' + 8y = e^{2t}$, $y(0) = 2$ & $y'(0) = -2$

(2) $y'' + 4y = \sin \omega t$; $y(0) = 0$ & $y'(0) = 0$

(3) $(D^2 + 6D + 9)x = 6t^2 e^{-3t}$; $x = 0$ & $Dx = 0$ at $t = 0$

(4) $x'' - 2x' + x = t^2 e^{-3t}$; $x(0) = 2$, $x'(0) = 3$

(5) $x'' + 3x' + 2x = 2(t^2 + t + 1)$; $x(0) = 2$, $x'(0) = 0$

(6) $(D^2 + 4D + 13)y = e^{-t} \sin t$; $y = 0$ & $Dy = 0$ at $t = 0$

(7) $y'' + y' - 2y = 3\cos 3t - 11 \sin 3t$; $y(0) = 0$ & $y'(0) = 6$

(8) $x'' + 9x = 18t$; $x(0) = 0$; $x(\pi/2) = 0$